$\Rightarrow \quad P = -2\,D$

$\therefore f = ?$

$\hookrightarrow \quad P = \frac{1}{f}$

$\Rightarrow -2 = \frac{1}{f}$

$\Rightarrow f = -\frac{1}{2}\,m.$

$= -\frac{1}{2} \times 100\,cm$

$\therefore f = -50\,cm.$

$\therefore$ The lens is a concave lens.

⇒ If a lens produced an image of an object which is 20 cm in size, then find the type of lens and the focal length, the object is at a distance of 25 cm.

[Given that ⇒ $m = -2$]

↳ $O = 20\,cm$ → useless

$u = -25\,cm$

$m = -2$

$f = ?$ $v = ?$

type of lens = ?

$m = \frac{v}{u}$

$-2 = \frac{v}{-25}$

$\therefore v = +50\,cm$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{f} = \frac{1}{50} - \frac{1}{(-25)}$

$= \frac{1+2}{50}$

$\Rightarrow \frac{1}{f} = \frac{3}{50}$

$\therefore f = \frac{50}{3} = 16.66\,cm.$

$\therefore$ focal length = 16.66 cm.

$\therefore$ It is convex lens.

A concave lens has a focal length of 20 cm. At what distance should a 5 cm tall object be placed in front of it so that it forms an image at 15 cm from the lens? Also, calculate the height of the image.

$\hookrightarrow f = -20\,cm$

$O = 5\,cm$

$u = ?$

$v = -15\,cm$

$I = ?$ [concave]

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\Rightarrow \frac{1}{-20} = \frac{1}{-15} - \frac{1}{u}$

$\Rightarrow -\frac{1}{20} + \frac{1}{15} = -\frac{1}{u}$

$\Rightarrow \frac{-3+4}{60} = \frac{1}{u}$

$\Rightarrow \frac{1}{60} = -\frac{1}{u}$

$\therefore u = -60\,cm.$

$m = \frac{I}{O} = \frac{v}{u}$

$\frac{I}{O} = \frac{v}{u}$

$\frac{I}{5} = \frac{-15}{-60}$

$\frac{I}{5} = \frac{1}{4}$

$I = \frac{5}{4} = 1.25$

$\therefore I = 1.25\,cm.$